

SAI RITHWIK AMAJALA

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EDUCATION

New York University — Tandon School of Engineering

Sep 2025 – May 2027

M.S., Computer Engineering | GPA: 3.833 | Coursework: Machine Learning (A), Deep Learning (A), ML Systems Engineering (A), Computing Systems Architecture (A), Real-Time Embedded Systems (A), High Performance ML

Vellore Institute of Technology (AP)

Aug 2020 – May 2024

B.Tech., Computer Science and Engineering | CGPA: 8.52/10 | Relevant: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Database Systems, Computer Architecture

TECHNICAL SKILLS

ML & Deep Learning: PyTorch (Distributed Data Parallel, DTensor, Fully Sharded Data Parallel, autograd, torch.compile), TensorFlow, scikit-learn, Hugging Face Transformers, PEFT / QLoRA, ONNX, TensorRT, inference optimization, quantization (INT8/INT4/NF4), distributed training (FSDP, NCCL), CUDA kernel engineering
Data & Analytics: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, PostgreSQL (query optimization, schema design), Redis, Elasticsearch, ETL pipeline design, data cleaning/validation, time-series analysis, statistical analysis, experiment tracking (MLflow)
Full-Stack & Cloud: FastAPI, Node.js, React, TypeScript, JavaScript, HTML/CSS, REST APIs, OAuth2/JWT/RBAC, Docker, Kubernetes, Terraform, AWS (EC2, S3, Lambda, SageMaker, CloudWatch), GCP, GitHub Actions (CI/CD), Prometheus, Grafana
Systems & Low-Level: C++ (multithreading, RAII, lock-free data structures), CUDA C/C++ (shared memory tiling, warp-level primitives), Nsight Systems/Compute profiling, Linux, Bash, Go
AI & Agents: Claude API, OpenAI API, Gemini API, LangChain (agent chains, tool orchestration, RAG, MCP), structured outputs / JSON schema, prompt engineering, guardrail design, LLM benchmarking

EXPERIENCE

Course Assistant — CS-GY 6923 Machine Learning, NYU Tandon School of Engineering

Sep 2026 – Present

- Supporting the graduate Machine Learning course (Cyber Fellows section, Prof. Regunathan Radhakrishnan): reviewing course material and preparing worked solutions to exercises covering classification, regression, SVMs, neural networks, clustering, dimensionality reduction, and model selection ahead of students.
- Set up and administer the course Slack workspace with structured channels for announcements, general discussion, and per-assignment Q&A; coordinating with the professor on course logistics.

Research Assistant — Pedestrian Attention & AV Safety, NYU HumanFUEL Lab (Prof. Linda Boyle)

May 2026 – Sep 2026

- Building and maintaining a gaze classification pipeline for the “So Pedestrian” IRB-approved study: processing 160K+ rows of eye-tracking data from Meta Project Aria smart glasses through ingestion, cleaning, validation, and feature engineering in Python (Pandas, NumPy); documenting methodology decisions, data schemas, and analysis protocols for reproducibility.
- Coordinating across hardware (wearable device calibration), software (gaze classification algorithms), and research design teams: translating technical requirements into structured project documentation, reporting weekly to PI Prof. Boyle and PhD supervisor Grace Douglas, and ensuring alignment with IRB-approved research protocol.

Computer Vision Member, NYU Ultraviolet Robotics Team (VIP)

Jan 2026 – Jun 2026

- Developing real-time perception systems for autonomous robots: YOLO detection + DeepSORT tracking on embedded GPU (Jetson) achieving sub-20ms p95 latency; TensorRT FP16 optimization; sensor fusion from camera, LiDAR, and IMU data via ROS; automated CI/CD regression testing reducing pipeline rebuild time by 3x.
- Assembling and maintaining robot hardware: wiring motor controllers, calibrating cameras and LiDAR sensors, debugging sensor-to-software integration; documenting system behavior and presenting weekly progress to faculty advisors and cross-disciplinary team members.

Open Source Contributor, gittuf — Software Supply Chain Security (Linux Foundation)

Oct 2025 – May 2026

- Contributing Go and Python improvements to production-grade security infrastructure: authentication pipeline hardening, trust policy verification, CI/CD automation, structured code reviews, and contributor onboarding documentation across a globally distributed team.

Software Engineer — Client: Synopsys, MulticoreWare India Pvt. Ltd.

May 2024 – Nov 2024

- Optimized ML inference pipelines across 5 workload families (classification, detection, segmentation, NLP, ASR) on GPU-accelerated hardware: operator fusion, subgraph partitioning, and mixed-precision quantization (FP32/FP16/INT8) on ONNX computation graphs achieving 22% p95 latency reduction and 35% GPU memory reduction across 40K+ benchmark runs.
- Built reproducible experiment infrastructure: automated ablation pipelines running 200+ configurations nightly, MLflow experiment tracking, Prometheus/Grafana dashboards for real-time latency monitoring (p50/p95/p99); caught 2 critical production regression bugs in quantized inference paths before deployment; reduced QA cycle time by 40%.
- Developed Python and Go backend services with REST API design, Docker containerization, and Kubernetes deployment across 3 staging environments; wrote comprehensive unit and integration tests (Pytest) achieving 95% coverage; participated in 30+ code reviews with an 8-person cross-functional engineering team.

SELECTED PROJECTS

Fused Linear Attention CUDA Kernel github.com/rithwik0908/fused-linear-attention

- Custom CUDA C++ kernel fusing QKV projection + attention + output into a single GPU pass: ~40% memory traffic reduction through shared memory tiling, warp-level shuffle reductions, and occupancy optimization; benchmarked against PyTorch and FlashAttention-2 across sequence lengths 512–16K with Nsight Compute roofline analysis.

HeatStop.AI — AI Agent Platform github.com/rithwik0908/HeatStop.AI

- End-to-end AI platform processing 50K+ records from 8 NYC data sources: Python ETL pipelines, PostgreSQL storage with query optimization, Redis caching delivering sub-100ms queries, Claude API + LangChain RAG with 4 deterministic guardrails reducing hallucination rate from 12% to <1%; FastAPI backend with OAuth2/JWT/RBAC; deployed on AWS (S3, CloudWatch, IAM).

CompanionBench — AI Evaluation Platform github.com/rithwik0908/CompanionBench

- Full-stack evaluation framework (Next.js/React, TypeScript, Prisma ORM, PostgreSQL, Node.js) comparing AI systems across 5 dimensions with automated browser testing (Playwright), real-time SSE streaming, and structured CSV/JSON export for stakeholder reporting; CI/CD deployment with Docker.

Distributed ML Training — 8x A100 Cluster github.com/rithwik0908

- Multi-node training using PyTorch Fully Sharded Data Parallel (FSDP) with NCCL: gradient compression reducing all-reduce volume by 30%, computation-communication overlap achieving 92% scaling efficiency; benchmarked DDP vs. FSDP vs. DTensor across 1B–13B parameter models.

LLM Fine-Tuning (QLoRA) github.com/rithwik0908/text-to-svg-qwen-competition

- Adapted Qwen2.5-1.5B with 4-bit NF4 QLoRA (LoRA rank 32, FP16, gradient checkpointing) on 12K samples: 75% model size reduction, <2% quality degradation; 5-step output validation pipeline reducing malformed outputs by 90%.

MLOps Serving Platform github.com/rithwik0908/mlops-serving

- Production infrastructure (Terraform, Kubernetes, Helm) with 5 services, MLflow model registry, Prometheus/Grafana monitoring; incident runbooks reducing MTTR from hours to <10 minutes.

PUBLICATIONS

Application of Advanced Data Mining and Computer Vision Techniques in License Plate Recognition | IEEE | DOI

Systematic comparison of YOLO, Faster R-CNN, SSD, OCR for real-time recognition systems.

Data-Driven Transformation of Hospitality Supply Chains Using AI-Powered Segmentation | IEEE | DOI

Applied K-Means, DBSCAN, agglomerative clustering, and PCA for operational segmentation and decision support.